# INORGANIC CUMULATIVE EXAM

## April 23, 1994

Attached are seven Tables and Figures (or parts thereof).

Choose five (5) of these and for each one you choose, write an explanation of the main idea(s) presented by the figure/table. Your answers will be graded with respect to the depth and breadth of your explanations.

1

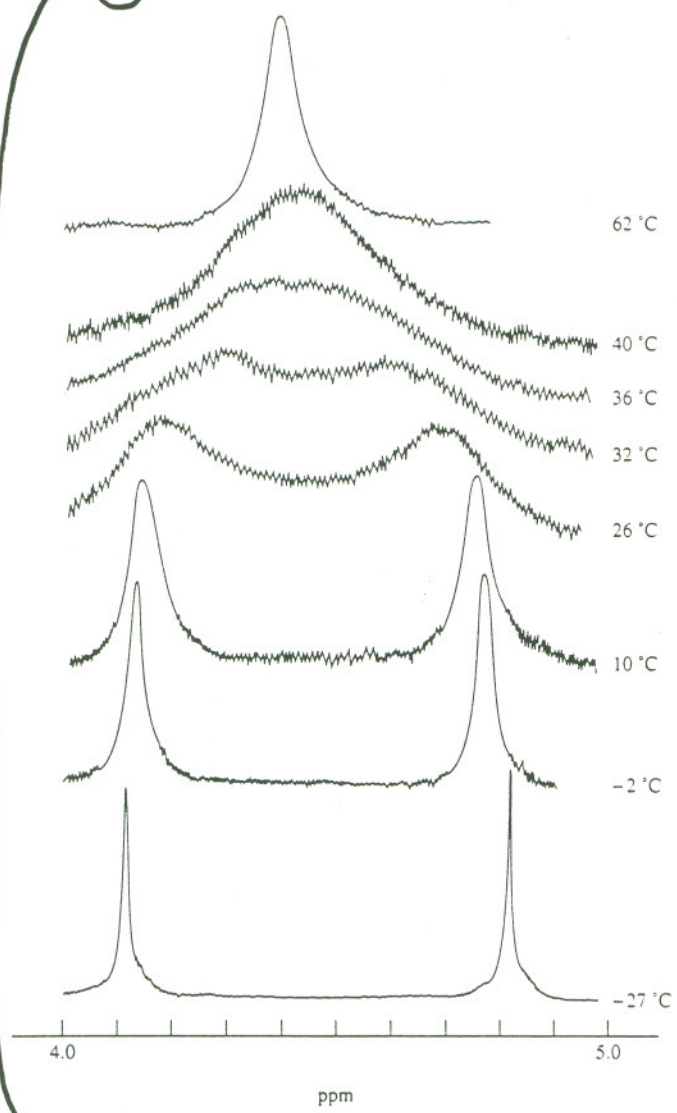



**Fig. 15.52** Proton NMR spectra of $(\eta^1\text{-}C_5H_5)_2(\eta^5\text{-}C_5H_5)_2Ti$ from −27 to +62 °C. The separate

2

**Table 13.4**

**Rate constants for the reaction of $[Co(NH_3)_5(H_2O)]^{3+}$ with $X^{n-}$ in water at 45 °C[a]**

| $X^{n-}$ | $k$ ($M^{-1}$ $s^{-1}$) |
|---|---|
| $NCS^-$ | $1.3 \times 10^{-6}$ |
| $H_2PO_4^-$ | $2.0 \times 10^{-6}$ |
| $Cl^-$ | $2.1 \times 10^{-6}$ |
| $NO_3^-$ | $2.3 \times 10^{-6}$ |
| $SO_4^{2-}$ | $1.5 \times 10^{-5}$ |

[a] Basolo, F.; Pearson, R. G. *Mechanisms of Inorganic Reactions*, 2nd ed.; Wiley: New York, 1967.

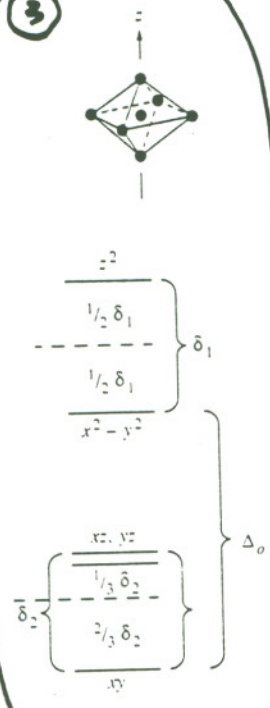
**Table 13.5**

**Rate constants for the reaction of $[Co(NH_3)_5X]^{m+}$ with $H_2O$[a]**

| $X^{n-}$ | $k$ ($s^{-1}$) |
|---|---|
| $NCS^-$ | $5.0 \times 10^{-10}$ |
| $H_2PO_4^-$ | $2.6 \times 10^{-7}$ |
| $Cl^-$ | $1.7 \times 10^{-6}$ |
| $NO_3^-$ | $2.7 \times 10^{-5}$ |
| $SO_4^{2-}$ | $1.2 \times 10^{-6}$ |

[a] Basolo, F.; Pearson, R. G. *Mechanisms of Inorganic Reactions*. 2nd ed.; Wiley: New York, 1967.

3



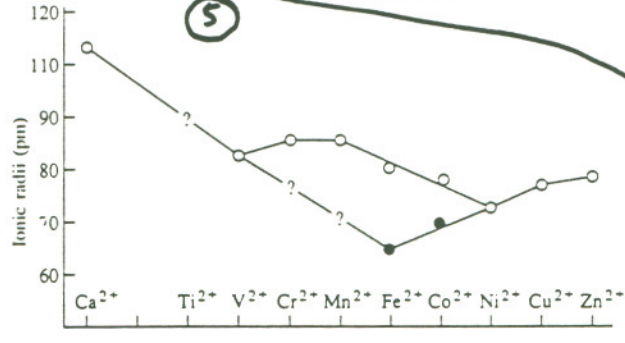

4

**Table 11.14**

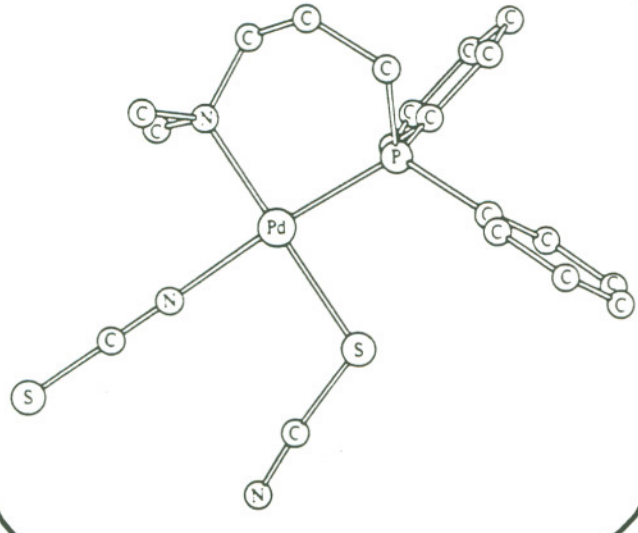
**Infrared carbonyl stretching frequencies ($cm^{-1}$) for some $W(CO)_5L$ complexes[a]**

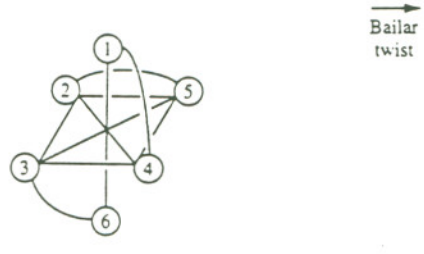
| L | $A_1^{(1)}$ |
|---|---|
| $Me_2NCHO$ | 1847 |
| $Me_2CO$ | 1847 |
| $Et_2O$ | 1908 |
| $H_2S$ | 1916 |
| $Ph_2S$ | 1930 |
| $Me_2S$ | 1932 |
| $CyNH_2$[d] | 1894 |
| pyridine | 1895 |
| aniline | 1916 |
| $CH_3CN$ | 1931 |
| $H_3P$ | 1921 |
| $Me_3P$ | 1947 |
| $Me_2PhP$ | 1947 |
| $MePh_2P$ | 1947 |
| $Ph_3P$ | 1942 |
| $(MeS)_3P$ | 1946 |
| $(EtO)_3P$ | 1959 |
| $(PhO)_3P$ | 1965 |
| $I_3P$ | 1981 |
| $Br_3P$ | 1991 |
| $Cl_3P$ | 1990 |
| $F_3P$ | 2007 |

5




6




7